AMD EPYC 9004 Series Processors - DATASHEET

Manufacturer: AMD
Part Number: EPYC-9654
Category: CPU / Server CPU
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Overview

The AMD EPYC 9654 is a 96-core, 192-thread server processor based on the Zen 4 architecture (Genoa). Base clock 2.4 GHz, boost up to 3.7 GHz. Built for cloud, enterprise virtualization and HPC workloads. Socket SP5, configurable TDP 320W to 400W (default 360W). Supports 1-socket and 2-socket configurations.

Memory and I/O

12-channel DDR5-4800 memory, up to 6 TB per socket. L3 cache: 384 MB. 128 lanes PCIe Gen5 (160 in 2P configurations). CXL 1.1+ support for memory expansion. Security: AMD Infinity Guard with SME, SEV-ES and SEV-SNP memory encryption. Up to 12 NUMA nodes per socket.

Key Specifications Summary

cores: 96. threads: 192. base: 2.4GHz. turbo: 3.7GHz. l3: 384MB. ddr5: 12ch. pcie5: 128lanes. tdp: 360W.